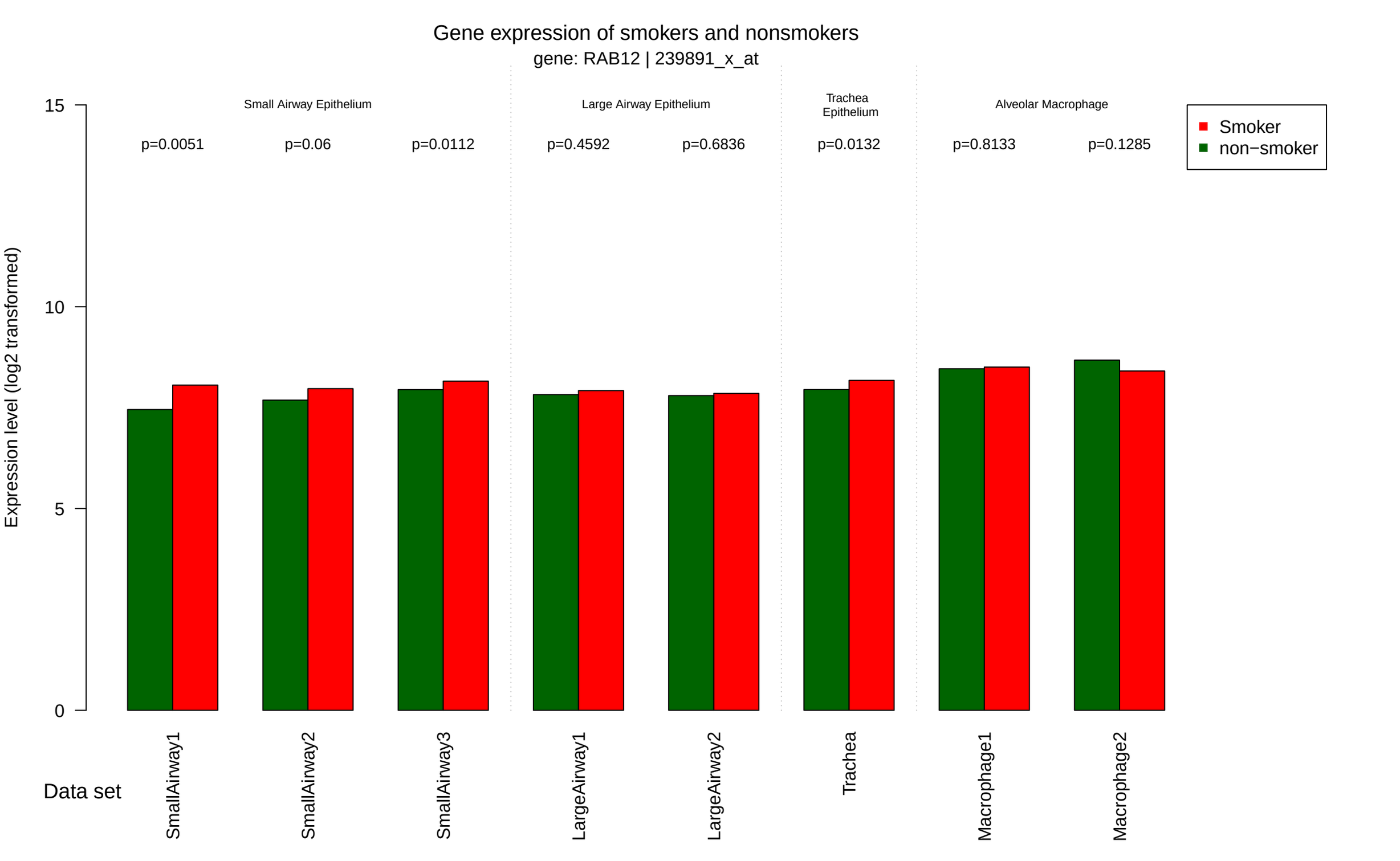
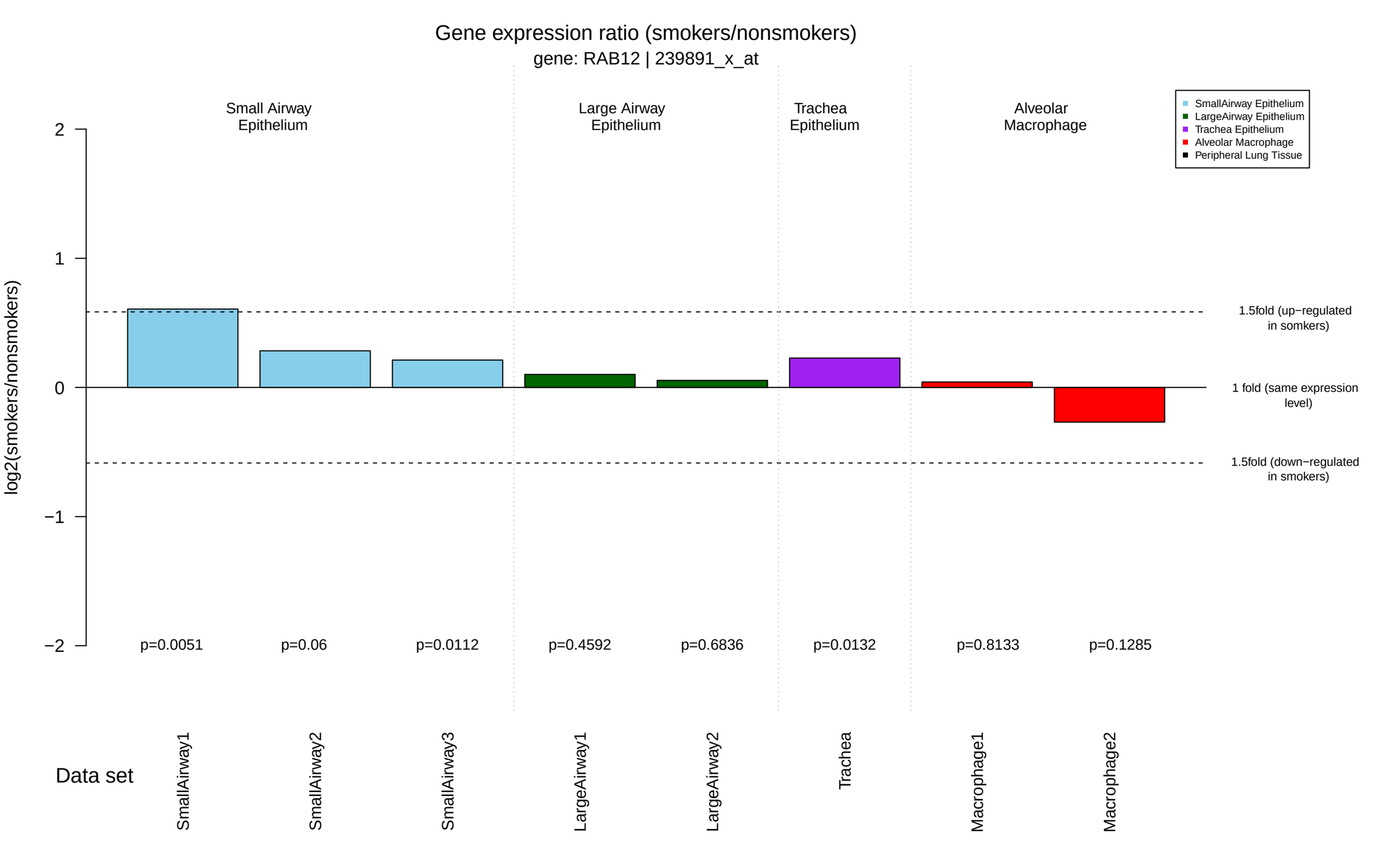
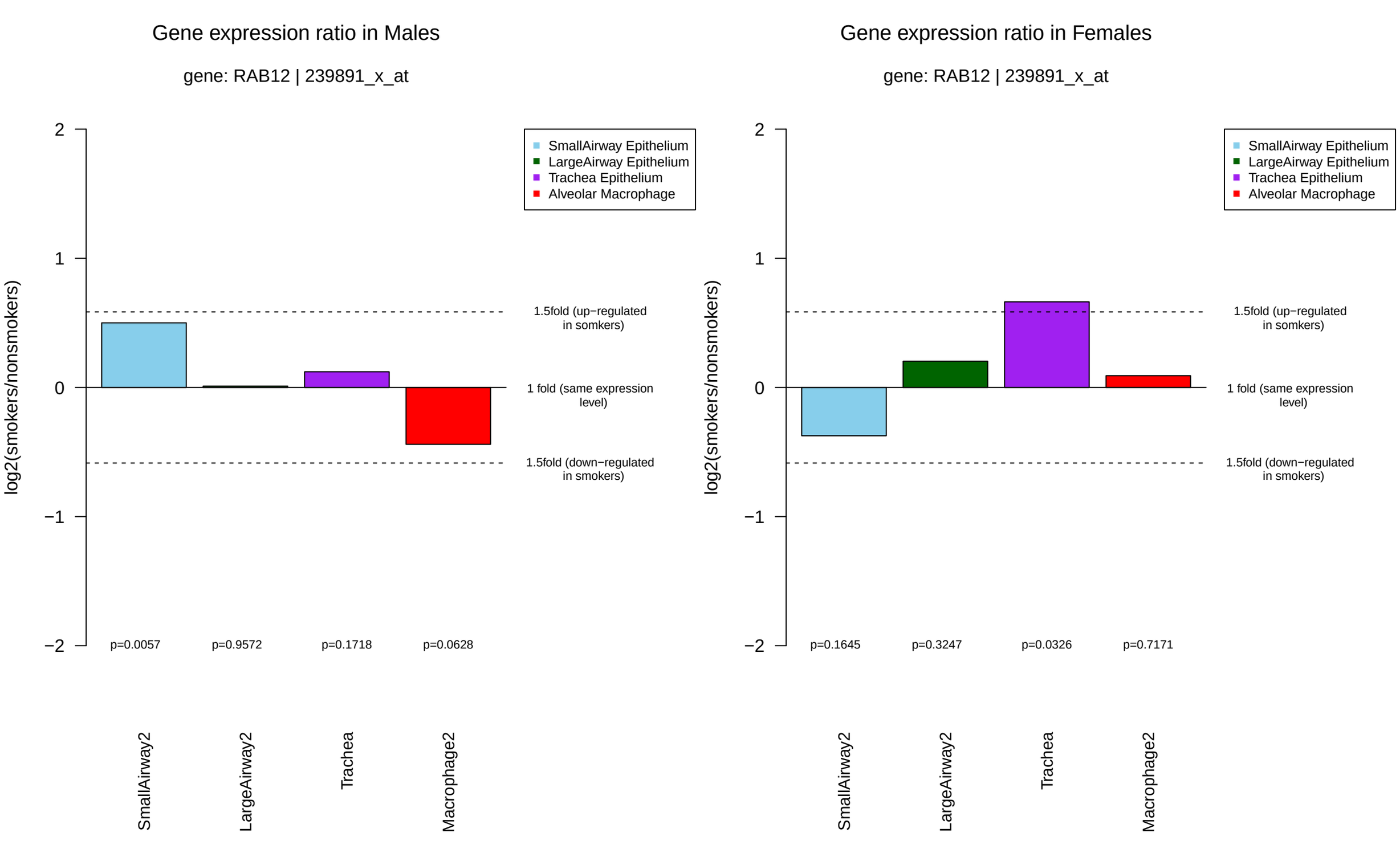
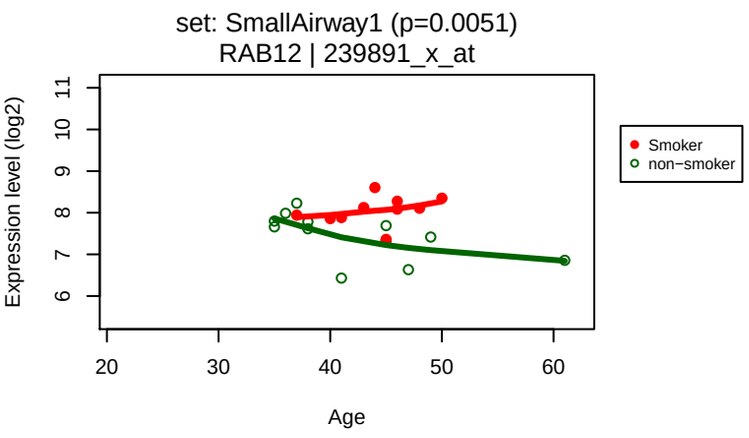
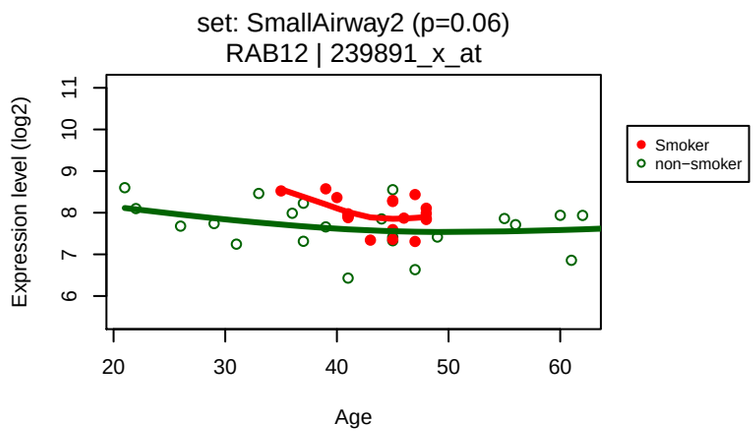
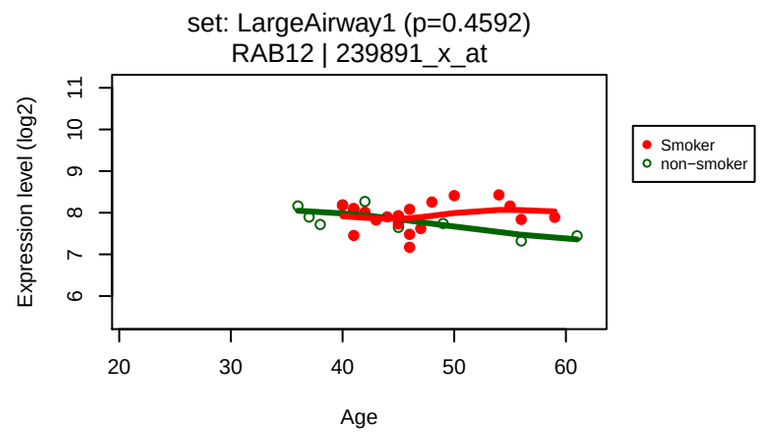

Gene expression of smokers and nonsmokers
gene: RAB12 | 239891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0051
p=0.06
p=0.0112
p=0.4592
p=0.6836
p=0.0132
p=0.8133
p=0.1285
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

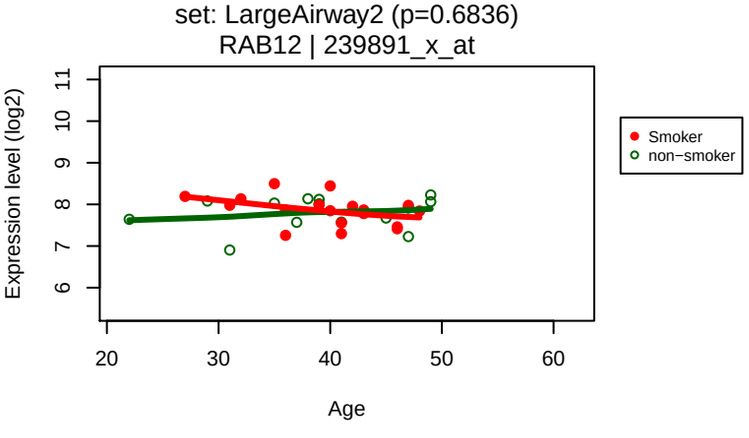

Gene expression ratio (smokers/nonsmokers)
gene: RAB12 | 239891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0051
p=0.06
p=0.0112
p=0.4592
p=0.6836
p=0.0132
p=0.8133
p=0.1285
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

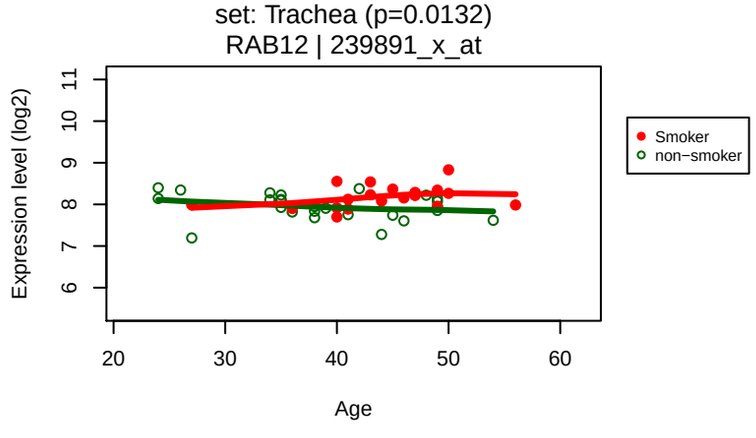

Gene expression ratio in Males
gene: RAB12 | 239891_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0057
p=0.9572
p=0.1718
p=0.0628
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB12 | 239891_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1645
p=0.3247
p=0.0326
p=0.7171
SmallAirway2
LargeAirway2
Trachea
Macrophage2

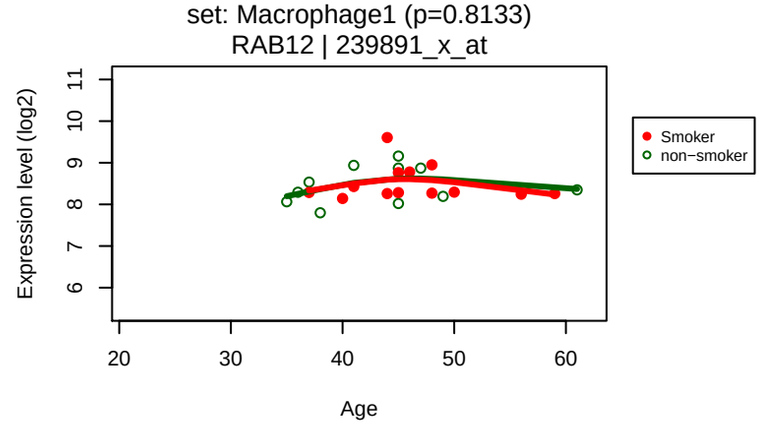

set: SmallAirway1 (p=0.0051)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker

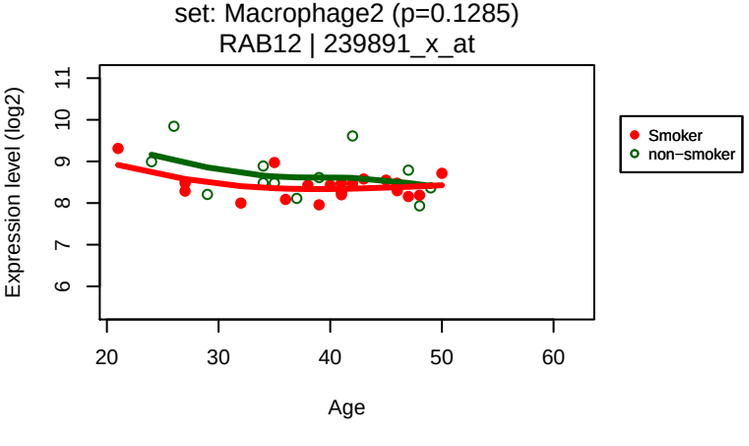

set: SmallAirway2 (p=0.06)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker

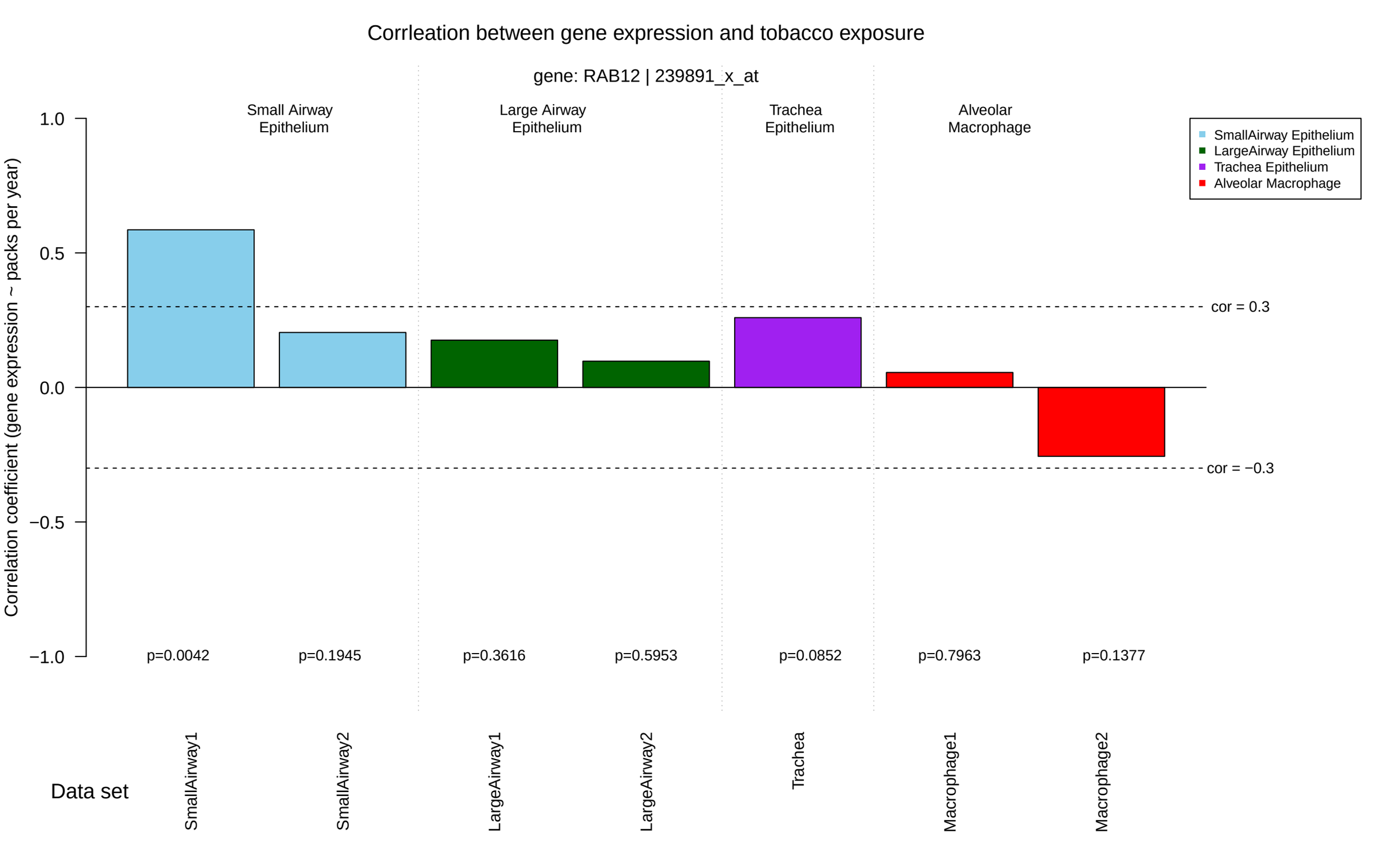

set: LargeAirway1 (p=0.4592)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6836)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0132)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8133)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1285)
RAB12 | 239891_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB12 | 239891_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0042
p=0.1945
p=0.3616
p=0.5953
p=0.0852
p=0.7963
p=0.1377
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2